

Getting started with AT32F426CBT7

Introduction

The AT-START-F426 evaluation board is designed to help you experience the high performance of the 32-bit microcontroller, the ARM Cortex®-M4F based AT32F426 series, and expedite development cycles and shorten time to market.

The AT-START-F426 evaluation board is based on the AT32F426CBT7 microcontroller. It features LEDs, buttons, one CANFD interface and Arduino™ Uno R3 extension connectors. It also comes with a built-in AT-Link-EZ, a tool designed to perform debugging/programming operations, without the need of other extra development tools.

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1 Overview

1.1 Features

The AT-START-F426 board has the following features:

- ARM Cortex®-M4F-based 32-bit AT32F426CBT7 microcontroller with FPU that embeds 128 KB Flash memory and 20 KB SRAM, in LQFP48 package
- On-board AT-Link connector:
 - The on-board AT-Link-EZ can be used for programming and debugging (AT-Link-EZ is a simplified version of AT-Link, and does not support offline mode)
 - If AT-Link-EZ is separated from this board by bending over along the joint, AT-START-F426 can be connected to an independent AT-Link for programming and debugging
- Power supply source:
 - USB bus of AT-Link-EZ
 - External 5 V power supply (E5V)
 - External 3.3 V power supply
- 4 x LED indicators:
 - LED1 (red) indicates that 3.3V power of the board is supplied
 - 3 x user LEDs, LED2 (red), LED3 (yellow) and LED4 (green) indicate operation status
- User button and Reset button
- 8 MHz HEXT crystal
- 32.768 kHz LEXT crystal
- Bus connector with a CANFD transceiver (up to 8 Mbit/s) to demonstrate that CAN1 supports CAN FD protocol
- Rich extension connectors:
 - Arduino™ Uno R3 extension connectors
 - LQFP48 I/O extension connectors

1.2 Definition of terms

- **Jumper JPx ON**
Jumped fitted
- **Jumper JPx OFF**
Jumped not fitted
- **Resistor Rx ON**
Short circuit by solder or 0 Ω resistor
- **Resistor Rx OFF**
Connections left Open

2 Quick start guide

2.1 Get started

Configure the AT-START-F426 board in the following sequence:

1. Check the Jumper position on board: JP1 is connected to GND
2. Connect the AT-Link-EZ to PC via USB cable (Type-A to Type-C) so that the board is powered via USB connector CN6. LED1 (red) is always on, and three other LEDs (LED2 to LED4) start to blink in turn.
3. After pressing User button (B2), the blinking frequency of three LEDs is changed.

2.2 Development toolchains

- ARM® Keil®: MDK-ARM™
- IAR™: EWARM
- AT32 IDE

3 Hardware layout and configuration

The AT-START-F426 board is designed around an AT32F426CBT7 microcontroller in LQFP48 package.

[Figure 1](#) shows the connection between AT-Link-EZ, AT32F426CBT7 and their peripherals (buttons, LEDs, a CANFD interface and extension connectors) on AT-START-F426 board.

[Figure 2](#) and [Figure 3](#) show their respective positions on the AT-Link-EZ and AT-START-F426 board.

Figure 1. Hardware block diagram

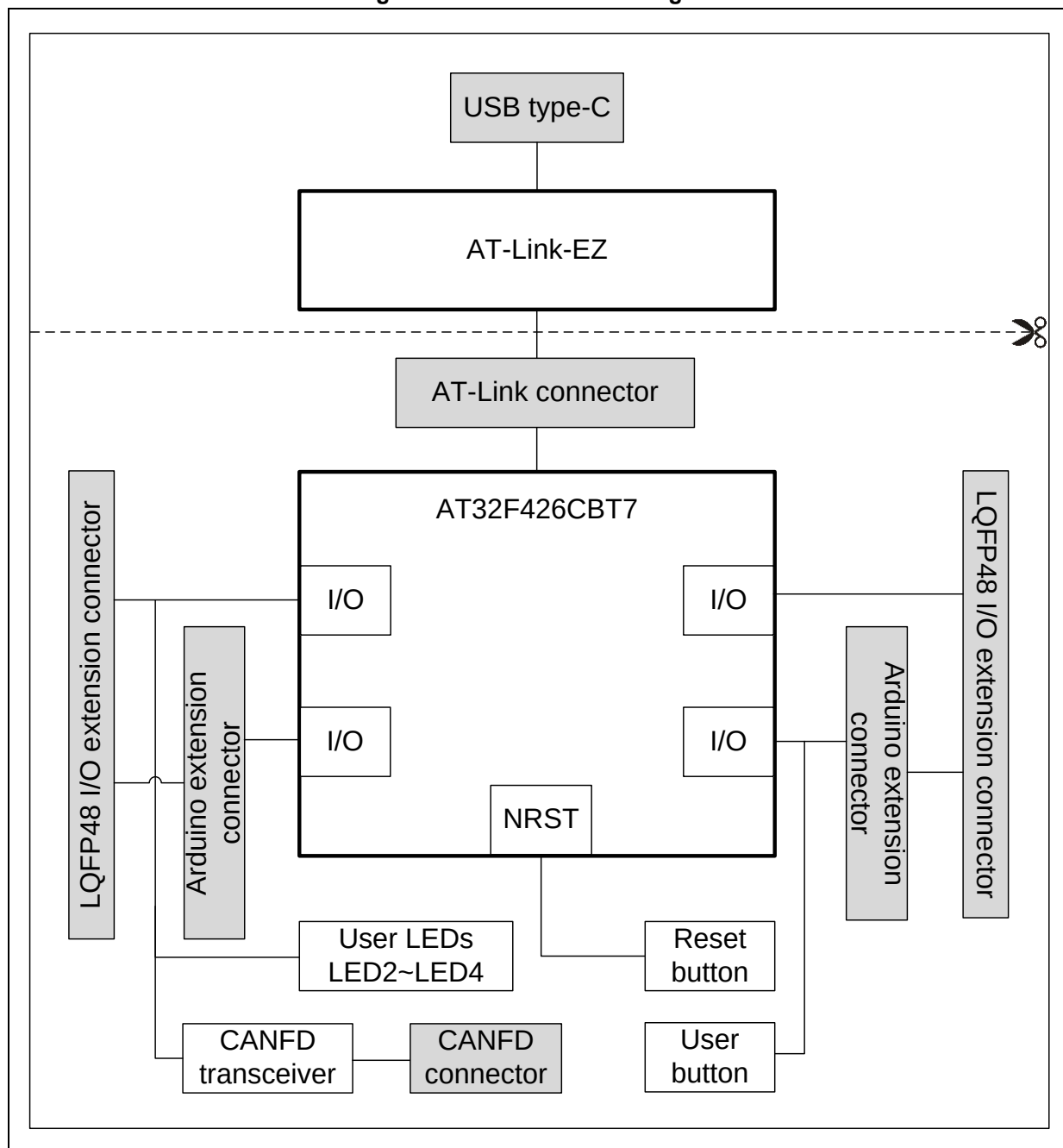


Figure 2. Top layer

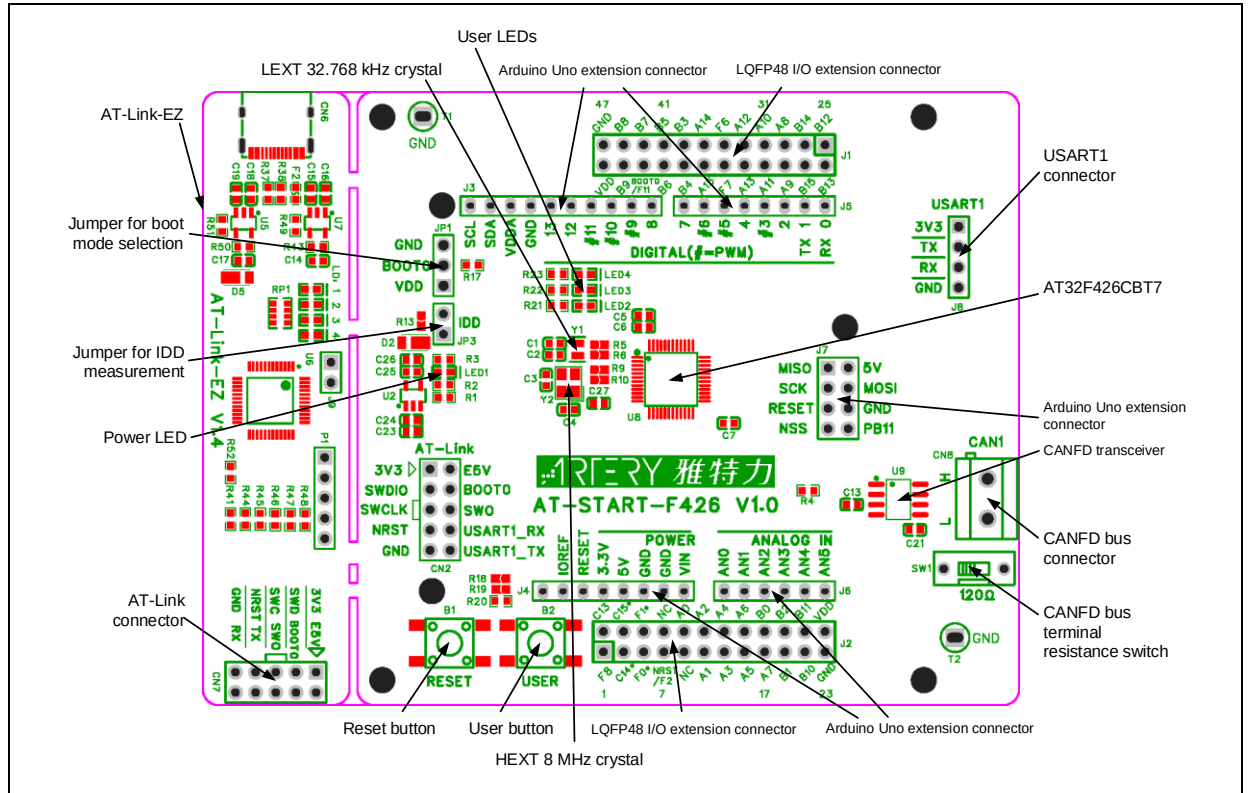
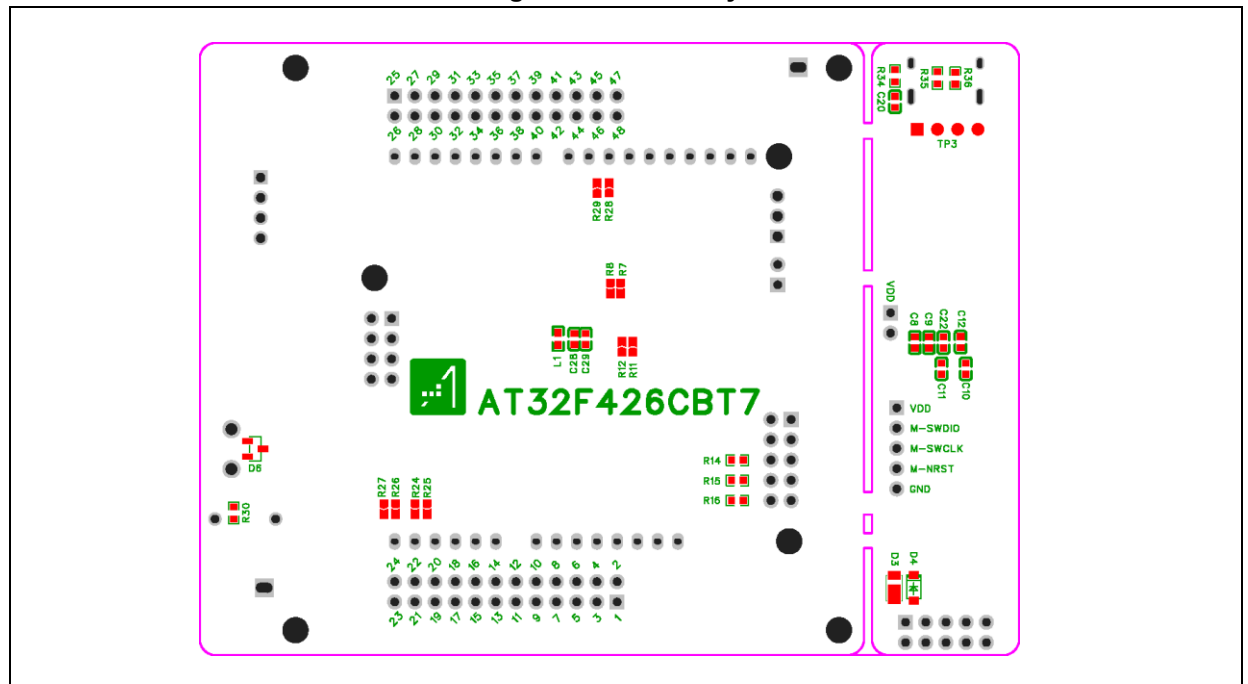


Figure 3. Bottom layer



3.1 Power supply sources

The 5 V power supply source required for AT-START-F426 can be provided from USB connector (CN6) of the AT-Link-EZ when the AT-Link-EZ is connected to PC via USB cable, or from an external 5 V power source (E5V). Then the 3.3 V required by the microcontroller and its peripherals is provided via 5 V-to-3.3 V voltage regulator (U2).

Additionally, the 5 V pin of J4 or J7 can be used as an input power source. Then the AT-START-F426 board must be powered by a 5 V power supply unit.

The 3.3 V pin of J4, or VDD pin of J1 and J2, can also be directly used as 3.3 V input power supply. The AT-START-F426 board must be powered by a 3.3 V power supply unit.

Note: Unless 5 V is provided via the USB connector (CN6) on the AT-Link-EZ, the AT-Link-EZ will not be powered by other power supply methods.

When another application board is connected to J4, the 5V and 3.3V can be used as output power supply, the 5V pin of J7 as 5 V output power supply, the VDD pin of J1 and J2 as 3.3 V output power supply.

3.2 IDD

When JP3 OFF (symbol IDD) and R13 OFF, an ammeter can be connected to measure the power consumption of AT32F426CBT7.

- **JP3 OFF, R13 ON**

AT32F426CBT7 is being powered (default setting, and JP3 connector is not mounted before shipping).

- **JP3 ON, R13 OFF**

AT32F426CBT7 is being powered.

- **JP3 OFF, R13 OFF**

An ammeter must be connected to measure the power consumption of AT32F426CBT7 (if no ammeter, the AT32F426CBT7 cannot be powered).

3.3 Embedded AT-Link-EZ for programming and debugging

The evaluation board integrates Artery AT-Link-EZ for users to program/debug the AT32F426CBT7 on the AT-START-F426 board. AT-Link-EZ supports SWD interface mode and SWO debugging. It also offers a virtual COM port (VCP) to be connected to the USART1_TX/USART1_RX (PA9/PA10) of the AT32F426CBT7.

Please refer to [AT-Link User Manual](#) for complete details on AT-Link-EZ.

The AT-Link-EZ can be separated from AT-START-F426 board. In this case, the CN2 connector (not mounted before shipping) of the AT-START-F426 board can be connected to the CN7 connector (not mounted before shipping) of the AT-Link-EZ or another AT-Link to reestablish connection, so as to continue programming and debugging the AT32F426CBT7.

3.4 Boot mode selection

At startup, the board boots from the following memory locations according to the BOOT0 setting. Users can modify the combination of nBOOT0_SW, nBOOT0 and nBOOT1 bits to select the desired boot mode, and the BOOT0 pin can be changed as PF11 functionality.

Table 1. Boot mode selection

Jumper	BOOT0 pin setting	Description
JP1 connected to GND	0	Boot from the internal Flash memory (Factory default settings)
JP1 connected to VDD	1	Boot from the system memory

3.5 External clock source

3.5.1 HEXT clock source

There are three hardware methods to configure the external high-speed clock sources:

- **On-board crystal (default setting)**

The 8 MHz crystal on board is used as HEXT clock source. Hardware settings: R9 and R10 must be ON, and R11, R12 OFF.

- **External oscillator from PF0**

External oscillator is from the pin 5 of J2. Hardware settings: R11 and R12 must be ON, and R9, R10 OFF.

- **HEXT not used**

PF0 and PF1 are used as GPIOs. Hardware settings: R11 and R12 must be ON, and R9, R10 OFF.

3.5.2 LEXT clock source

There are three hardware methods to configure the external low-speed clock sources:

- **On-board crystal (default setting)**

The 32.768 kHz crystal on board is used as LEXT clock source. Hardware settings: R5 and R6 must be ON, and R7, R8 OFF.

- **External oscillator from PC14**

External oscillator is from the pin 3 of J2. Hardware settings: R7 and R8 must be ON, and R5, R6 OFF.

- **LEXT not used**

PC14 and PC15 are used as GPIOs. Hardware settings: R7 and R8 must be ON, and R5, R6 OFF.

3.6 LEDs

- **Power LED1**

Red, indicating that the AT-START-F426 board is powered by 3.3 V.

- **User LED2**

Red, controlled with the PF6 pin of AT32F426CBT7. The User LED lights on when PF6 outputs low.

- **User LED3**

Yellow, controlled with the PF7 pin of AT32F426CBT7. The User LED lights on when PF7 outputs low.

- **User LED4**

Green, controlled with the PF8 pin of AT32F426CBT7. The User LED lights on when PF8 outputs low.

3.7 Buttons

- **Reset button B1**

It is connected to NRST/PF2 (used as NRST by default) to reset AT32F426CBT7. It can be changed as PF2 functionality by setting the NRST_MODE bit in User System Data.

- **User button B2**

By default, it is connected to the PA0 of AT32F426CBT7 and used as a WKUP1/TAMP2 button (R18 ON, R19 OFF); it can also be connected to PC13 and used as a WKUP2/TAMP1 button (R18 OFF, R19 ON).

3.8 CANFD communication

The CAN1 connector on AT-START-F426 board is connected to a transceiver for CANFD communication. The transceiver MCP2562FD-E/SN (U9) supports up to 8 Mbit/s of communication speed. CANH/CANL bus offers connectors that can be connected externally through bolt-type terminal block (CN8). A 120 Ω terminal resistor for CAN1 bus at the board end can be selected by the SW1 switch. When the SW1 switch is ON, the terminal resistor is connected to CANH/CANL bus. The MCP2562FD-E/SN chip STBY pin is connected to the AT32F426CBT7 pin CAN1_STB (PB7) pin. When the STBY pin is high, the MCP2562FD-E/SN can be set in Standby mode.

3.9 0 Ω resistors

Table 2. 0 Ω resistor settings

Resistors	State ⁽¹⁾	Description
R13 (microcontroller power consumption measurement)	ON	When JP3 OFF, the microcontroller is powered by 3.3 V.
	OFF	When JP3 OFF, an ammeter can be connected to 3.3 V to measure the power consumption of microcontroller (if no ammeter, the microcontroller cannot be powered).
R9, R10, R11, R12 (HEXT)	ON, ON, OFF, OFF	The crystal Y2 on board is used as HEXT clock source.
	OFF, OFF, ON, ON	HEXT clock source is from external PF0, or PF0 and PF are used as GPIOs.
R5, R6, R7, R8 (LEXT)	ON, ON, OFF, OFF	The crystal Y1 on board is used as LEXT clock source.
	OFF, OFF, ON, ON	LEXT clock source is from external PC14, or PC14 and PC15 are used as GPIOs.

Resistors	State ⁽¹⁾	Description
R18, R19 (User button B2)	ON, OFF	User button B2 is connected to PA0.
	OFF, ON	User button B2 is connected to PC13.
R24, R25, R26, R27 (Arduino™ AN4, AN5)	OFF, ON, OFF, ON	Arduino™ AN4 and AN5 are connected to ADC1_IN6 and ADC1_IN9.
	ON, OFF, ON, OFF	Arduino™ AN4 and AN5 are connected to I2C1_SDA and I2C1_SCL.
R28, R29 (Arduino™ D10)	OFF, ON	Arduino™ D10 is connected to SPI1_CS.
	ON, OFF	Arduino™ D10 is connected to PWM (TMR4_CH1).

(1) The factory default Rx state is shown in **BOLD** font.

3.10 Extension connectors

3.10.1 Arduino™ Uno R3 extension connectors

Female connectors J3~J6 and male J7 support standard Arduino™ Uno R3 connectors. Most of the daughter boards designed around Arduino™ Uno R3 are fit for AT-START-F426.

Note 1: The I/O ports of AT32F426CBT7 are 3.3 V compatible with Arduino™ Uno R3, but 5 V not.

Note 2: The pin 8 of J3 is VDDA, without AREF function of Arduino™ Uno R3.

Table 3. Arduino™ Uno R3 extension connectors

Connectors	Pin No.	Arduino pin name	AT32F426 pin name	Function
J4 (Power supply)	1	NC	-	-
	2	IOREF	-	3.3 V reference voltage
	3	RESET	NRST/PF2	External reset
	4	3.3V	-	3.3 V input/output
	5	5V	-	5 V input/output
	6	GND	-	Ground
	7	GND	-	Ground
	8	NC	-	-
J6 (Analog input)	1	AN0	PA0	ADC1_IN0
	2	AN1	PA1	ADC1_IN1
	3	AN2	PA4	ADC1_IN4
	4	AN3	PB0	ADC1_IN8
	5	AN4	PA6 or PB9 ⁽¹⁾	ADC1_IN6 or I2C1_SDA
	6	AN5	PB1 or PB8 ⁽¹⁾	ADC1_IN9 or I2C1_SCL
J5 (Logic input/output low byte)	1	D0	PA3	USART2_RX
	2	D1	PA2	USART2_TX
	3	D2	PA10	-
	4	D3	PB5	TMR3_CH2
	5	D4	PB3	-
	6	D5	PB4	TMR3_CH1
	7	D6	PA8	TMR1_CH1
	8	D7	PB10	-

Connectors	Pin No.	Arduino pin name	AT32F426 pin name	Function
J3 (Logic input/output high byte)	1	D8	PA9	-
	2	D9	PB7	TMR4_CH2
	3	D10	PA15 or PB6 ⁽¹⁾	SPI1_CS or TMR4_CH1
	4	D11	PA7	TMR3_CH2 / SPI1_MOSI
	5	D12	PA6	SPI1_MISO
	6	D13	PA5	SPI1_SCK
	7	GND	-	Ground
	8	VDD	-	3.3 V input/output
	9	SDA	PB9	I2C1_SDA
	10	SCL	PB8	I2C1_SCL
J7 (Others)	1	MISO	PB14	SPI2_MISO
	2	E5V	-	5 V input/output
	3	SCK	PB13	SPI2_SCK
	4	MOSI	PB15	SPI2_MOSI
	5	RESET	NRST/PF2	External reset
	6	GND	-	Ground
	7	NSS	PB12	SPI2_CS
	8	GPIO	PB11	-

(1) 0 Ω resistor settings are shown in [Table 2](#).

3.10.2 LQFP48 I/O extension connectors

The extension connectors J1 and J2 are used to connect the IO ports of the AT-START-F426 to external devices. All the I/O ports of AT32F426CBT7 are accessible. J1 and J2 can also be measured with oscilloscope, logic analyzer or voltmeter probe.

4 Revision history

Table 4. Document revision history

Date	Version	Revision note
2025.7.4	1.00	Initial release.

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